

ML Interconnection Queue Intelligence for the U.S. Electric Grid

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Abstract

The U.S. electric grid faces a growing interconnection bottleneck: thousands of gigawatts of proposed generation and storage capacity are waiting in interconnection queues, yet most projects are eventually withdrawn before reaching operation. Developers must make expensive siting, permitting, and financing decisions before receiving reliable information about queue survival, study duration, or network upgrade costs. This project develops a machine-learning pipeline for early-stage interconnection queue intelligence. The system predicts three project-level outcomes: withdrawal risk, interconnection study timeline, and network upgrade cost severity. The methodological contribution is an adaptation of the physics-informed neural network (PINN) framework of Raissi, Perdikaris, and Karniadakis [1]. Instead of enforcing a partial differential equation residual, the cost model adds a domain-informed monotonicity residual: holding queue context fixed, increasing requested MW capacity should not decrease the predicted upgrade-cost bucket. The final pipeline includes interpretable linear baselines, neural-network models, a PINN-style monotonic cost classifier, and Gaussian process regression for cost uncertainty. On the reproducible project dataset, withdrawal prediction achieved AUC-ROC of 0.949 for logistic regression and 0.946 for the MLP classifier; study timeline prediction achieved 4.06 months MAE with Ridge regression; the cost-bucket MLP achieved weighted F1 of 0.887; and Gaussian process regression achieved 88.6% empirical coverage for nominal 90% prediction intervals. The results show that queue records contain useful signal for early-stage risk screening, while also highlighting the limits imposed by incomplete cost labels, simplified physics, and public-data inconsistencies.

1 Introduction

Connecting new generation to the U.S. electric grid requires passing through an interconnection queue. Before a solar plant, wind farm, battery project, hybrid facility, or conventional generator can operate, the relevant transmission provider or ISO/RTO must study whether the grid can accommodate the project and what network upgrades are needed. This process is technically necessary: interconnection studies protect reliability by checking thermal limits, voltage behavior, short-circuit duty, deliverability, and the effect of new injections on existing network constraints. However, the process has become one of the main bottlenecks in the clean-energy transition.

Recent public reports from Lawrence Berkeley National Laboratory (LBNL) describe an interconnection system with unprecedented volume and attrition. More than 2,000 GW of proposed generation and storage capacity has been waiting in U.S. queues, and a large majority of projects historically withdraw before reaching commercial operation [2]. Developers therefore often make expensive siting, land, permitting, and financing decisions before they know whether a project will

survive the queue or whether network upgrades will make it uneconomic. A better early-stage estimate of queue risk could reduce wasted development effort and support better project siting.

This project frames interconnection queue intelligence as a project-level machine learning problem. Given information available near the time of filing, the goal is to predict three outcomes:

1. Withdrawal risk: the probability that a project eventually withdraws from the interconnection queue.
2. Study timeline: the number of months required to complete the interconnection study process.
3. Network upgrade cost severity: a categorical estimate of required network upgrade cost, bucketed as \$0–1M, \$1–5M, \$5–20M, and \$20M+.

The work follows the implementation and demo format of the course project. It also connects directly to the physics-informed machine learning topic covered in CEE 498 MLC. The research paper reproduced at the methodological level is Raissi, Perdikaris, and Karniadakis (2019), “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations” [1]. The original paper trains neural networks using a composite objective consisting of a data loss and a physics residual. In the original setting, the residual is derived from a governing PDE and evaluated at collocation points. In this project, public queue data is tabular and does not expose the full grid state needed to enforce AC power-flow equations directly. I therefore adapt the same composite-loss structure by replacing the PDE residual with a grid-domain monotonicity residual for cost prediction.

2 Related Work and Research Paper Reproduction

2.1 Physics-Informed Neural Networks

Raissi et al. [1] introduced physics-informed neural networks as a framework for training neural networks to satisfy both observed data and known physical laws. The general continuous-time formulation assumes a nonlinear PDE of the form

$$u_t + \mathcal{N}[u] = 0, \tag{1}$$

where u is the unknown solution field and $\mathcal{N}[\cdot]$ is a nonlinear differential operator. A neural network approximates $u(t, x)$, and automatic differentiation is used to compute the PDE residual

$$f(t, x) = u_t + \mathcal{N}[u]. \tag{2}$$

Training minimizes a composite objective,

$$L(\theta) = L_{\text{data}}(\theta) + \lambda L_{\text{physics}}(\theta), \tag{3}$$

where L_{data} fits observed data and L_{physics} penalizes nonzero PDE residuals at collocation points. In the paper, this approach is demonstrated on Burgers’ equation, Schrödinger dynamics, Allen–Cahn systems, Navier–Stokes flow, and Korteweg–de Vries dynamics. The physics residual acts as a regularizer, especially when data are sparse.

2.2 Adapting the PINN Idea to Queue Data

Interconnection queue records are not a continuous spatiotemporal field. They are project-level records with features such as capacity, fuel type, ISO/RTO region, filing year, voltage, queue position, and point-of-interconnection context. Public queue exports also do not include enough physical grid state to compute full AC power-flow residuals: line impedances, topology, voltage magnitudes, generation dispatch, contingency lists, thermal limits, and affected-system constraints are generally not available in a project-level queue record.

For this reason, the project reproduces the composite-loss framework rather than a literal PDE experiment. The domain residual encodes a simplified but meaningful interconnection principle: holding project context fixed, increasing requested MW capacity should not reduce expected network upgrade cost severity. Larger injections tend to require at least as much transfer capability and can trigger larger thermal, deliverability, or voltage-related upgrades. This is not a complete power-flow model, but it is a grid-informed monotonicity prior that can regularize the cost classifier in sparse or noisy regions.

3 Dataset and Feature Engineering

3.1 Data Sources and Reproducible Dataset

The intended public-data backbone is the LBNL Queued Up dataset, which compiles project-level interconnection queue records across U.S. ISO/RTOs and non-ISO utilities representing most installed U.S. generating capacity [2]. The project proposal also identified supplementary ISO/RTO queue exports, especially MISO, PJM, and CAISO public queue reports, for fields such as substation, county, voltage, study phase, queue position, and deliverability status.

The repository includes a runnable data preparation path and a proposal-scale synthetic fallback dataset generator. The fallback does not replace the real-data motivation. Instead, it preserves the schema, class imbalance, missing-cost-label pattern, and qualitative relationships described in the proposal and progress report so that the full pipeline is executable and verifiable. The same loader can ingest a cleaned CSV or Excel file with real queue records using the `-data` argument.

The final reproducible run used 12,000 project records. The train/validation/test split was 80/10/10, producing 9,600 training rows, 1,200 validation rows, and 1,200 test rows. Cost labels were available for 4,581 records, corresponding to 38.2% cost-label coverage. The withdrawal rate was 77.25%, reflecting the strong imbalance reported in public queue literature.

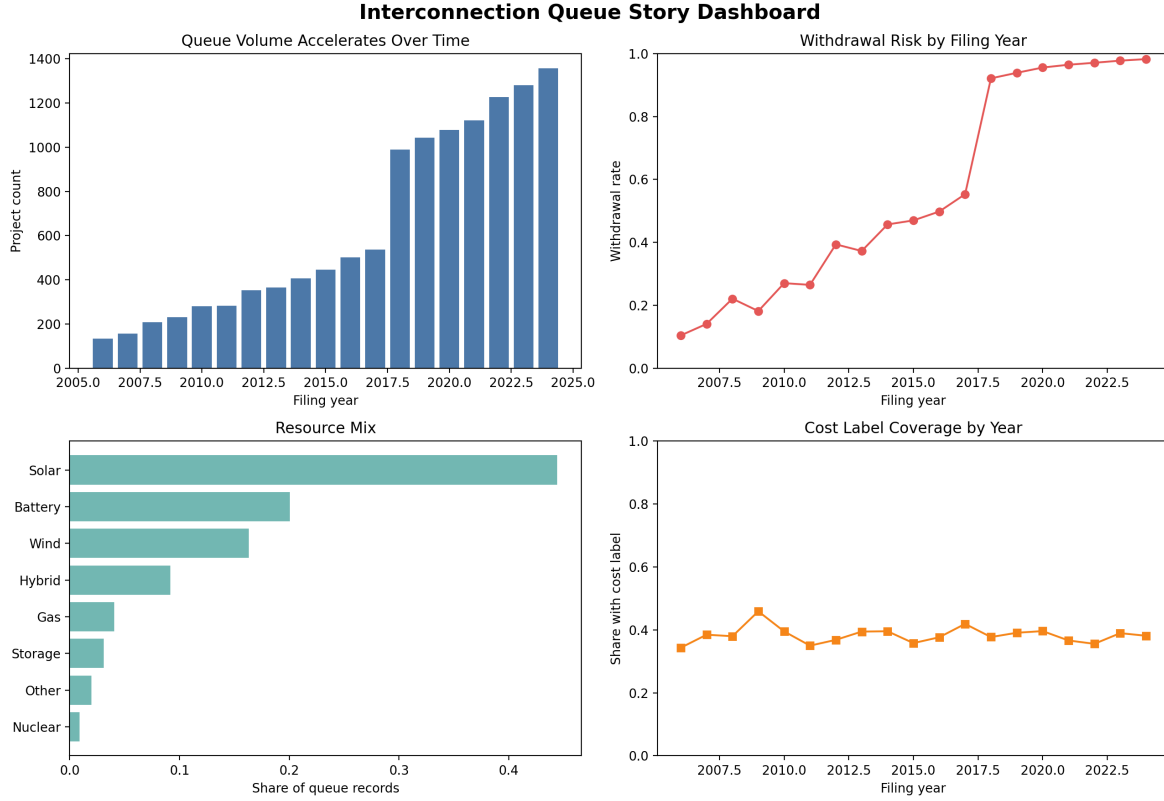


Figure 1: Exploratory dashboard for the reproducible interconnection queue dataset. Queue entry volume increases sharply in later filing years, while withdrawal and cost-label patterns remain central modeling challenges.

3.2 Feature Set

The feature set combines project characteristics, queue timing, and local grid-context proxies. Numeric variables are standardized using statistics fit on the training split only. Categorical variables are one-hot encoded with unknown categories ignored at test time.

Table 1: Feature set used in the project pipeline.

Feature	Description
log_mw_capacity	Log-transformed requested project capacity in MW.
iso_region	ISO/RTO or regional queue identifier.
fuel_type	Resource type, such as solar, wind, battery, hybrid, gas, or other.
filing_year	Year the project entered the queue.
voltage_kv	Interconnection voltage level.
queue_position	Project position or queue-order proxy.
substation_congestion_24mo	Count of prior projects at the same point of interconnection within the previous 24 months.
queue_cohort_size	Number of projects entering the same ISO/year cohort.
county_queue_density	Number of projects in the same county/year cohort.
distance_to_transmission_miles	Siting proxy for distance to nearby transmission.
renewable_penetration_pct	Regional renewable saturation proxy.
deliverability_status	Requested deliverability product, such as energy-only or full-capacity deliverability.
interconnection_service	Interconnection service type, such as ERIS, NRIS, capacity resource, or network resource.

The most important engineered feature is the substation congestion proxy. It counts other projects queued at the same point of interconnection in the preceding two years. This feature captures the intuition that a project connecting at an already congested location is more likely to face upgrade requirements, delays, and adverse economic outcomes.

4 Methodology

4.1 Overall Pipeline

The pipeline contains four modeling components: withdrawal prediction, timeline prediction, cost-bucket classification, and cost uncertainty estimation. Raw project records are transformed into a standardized feature vector. Separate model heads are trained for each prediction task. The PINN-style residual is applied only to the cost-bucket neural network because the monotonicity rule concerns network upgrade cost severity.

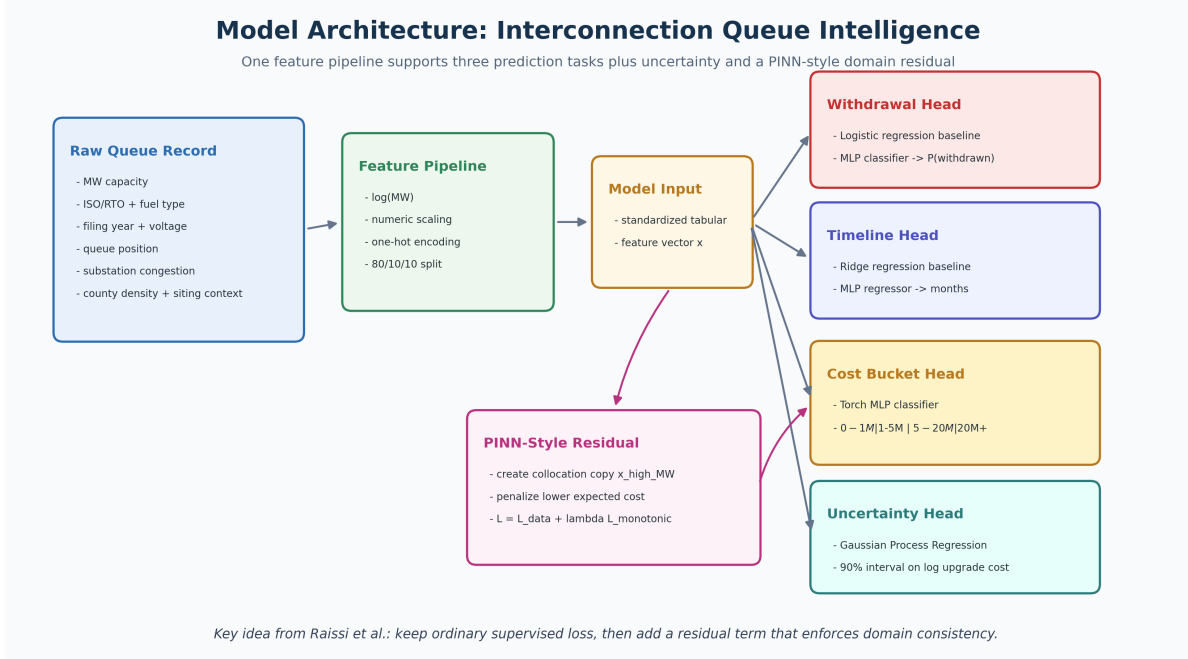


Figure 2: Model architecture. A shared feature pipeline supports separate prediction heads for withdrawal, timeline, cost bucket, and cost uncertainty. The PINN-style residual regularizes the cost classifier using higher-MW collocation copies.

4.2 Withdrawal Prediction

Withdrawal is modeled as binary classification:

$$y_i^{\text{withdraw}} \in \{0, 1\}. \quad (4)$$

Two models are compared. The baseline is logistic regression with class weighting to account for withdrawal imbalance. The nonlinear model is a multilayer perceptron (MLP) classifier with two hidden layers, early stopping, and L_2 regularization. The key metrics are AUC-ROC, F1 score on the withdrawn class, and accuracy.

4.3 Study Timeline Prediction

Study timeline is modeled as regression:

$$y_i^{\text{timeline}} \in \mathbb{R}_+, \quad (5)$$

where the target is the number of months required to complete the interconnection study process. The baseline is Ridge regression, and the nonlinear comparison model is an MLP regressor. The main metrics are mean absolute error (MAE) in months and coefficient of determination R^2 .

4.4 Cost-Bucket Classification

Network upgrade cost is modeled as a four-class classification problem on the subset with cost labels. The ordered cost buckets are

$$\{\$0-1M, \$1-5M, \$5-20M, \$20M+\}.$$

The unconstrained model is a Torch MLP classifier trained with cross-entropy:

$$L_{\text{cost}}(\theta) = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^4 \mathbb{I}[y_i = k] \log p_{\theta}(y_i = k \mid x_i). \quad (6)$$

The physics-informed version uses the same architecture and data but adds the monotonic residual. Let x_i be the original project record. A collocation copy x_i^+ is formed by increasing requested MW capacity while holding all other features fixed. Let $s_{\theta}(x)$ be the expected cost bucket index under the model’s softmax output:

$$s_{\theta}(x) = \sum_{k=0}^3 k p_{\theta}(y = k \mid x). \quad (7)$$

The monotonicity penalty is

$$L_{\text{mono}}(\theta) = \frac{1}{N} \sum_{i=1}^N \max(0, s_{\theta}(x_i) - s_{\theta}(x_i^+)). \quad (8)$$

If the expected cost bucket decreases when MW increases, the penalty is positive. If the expected cost remains the same or increases, the penalty is zero. The constrained training objective is

$$L_{\text{PINN}}(\theta) = L_{\text{cost}}(\theta) + \lambda L_{\text{mono}}(\theta). \quad (9)$$

This is the direct analogue of Equation 3: the PDE residual from Raissi et al. is replaced by a domain residual appropriate for tabular queue data.

4.5 Gaussian Process Cost Uncertainty

Gaussian process regression (GPR) is used to estimate uncertainty for network upgrade cost on the cost-labeled subset. The target is log-transformed cost:

$$z_i = \log(1 + \text{cost}_{i,\text{MUSD}}). \quad (10)$$

The model uses a radial basis function kernel with a constant scale and a white-noise term:

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right) + \sigma_n^2 \delta_{x, x'}. \quad (11)$$

GPR returns both a posterior mean and a posterior standard deviation. A nominal 90% prediction interval is computed as

$$\hat{\mu}(x) \pm 1.645 \hat{\sigma}(x). \quad (12)$$

5 Experimental Setup

5.1 Train/Validation/Test Split

The dataset is split into 80% training, 10% validation, and 10% test. Where feasible, the split is stratified by ISO region and filing-year cohort so that each split contains similar temporal and regional structure. All preprocessing steps are fit on the training data only. Test metrics are computed once on the held-out test set.

Cost-bucket classification and GPR are trained only on the cost-labeled subset. This is important because network upgrade cost labels are not available for every queue project.

Table 2: Evaluation metrics by task.

Task	Metrics
Withdrawal prediction	AUC-ROC, F1 score for withdrawn class, accuracy.
Timeline prediction	MAE in months, R^2 .
Cost-bucket classification	Weighted F1, accuracy, monotonic implausibility rate.
GPR uncertainty	MAE on log cost, R^2 on log cost, 90% PICP, mean interval width.

6 Results

6.1 Withdrawal Prediction

Table 3: Withdrawal prediction results on the held-out test set.

Model	AUC-ROC	F1 (withdrawn)	Accuracy
Logistic regression	0.949	0.908	0.868
MLP classifier	0.946	0.932	0.895

Both models perform substantially above random ranking and majority-class baselines. Logistic regression achieves the best AUC-ROC, while the MLP classifier achieves the best F1 and accuracy. Figure 3 provides ROC, precision–recall, calibration, and confusion-matrix diagnostics. Figure 4 shows a risk surface over requested MW and prior substation congestion while holding other features at representative values.

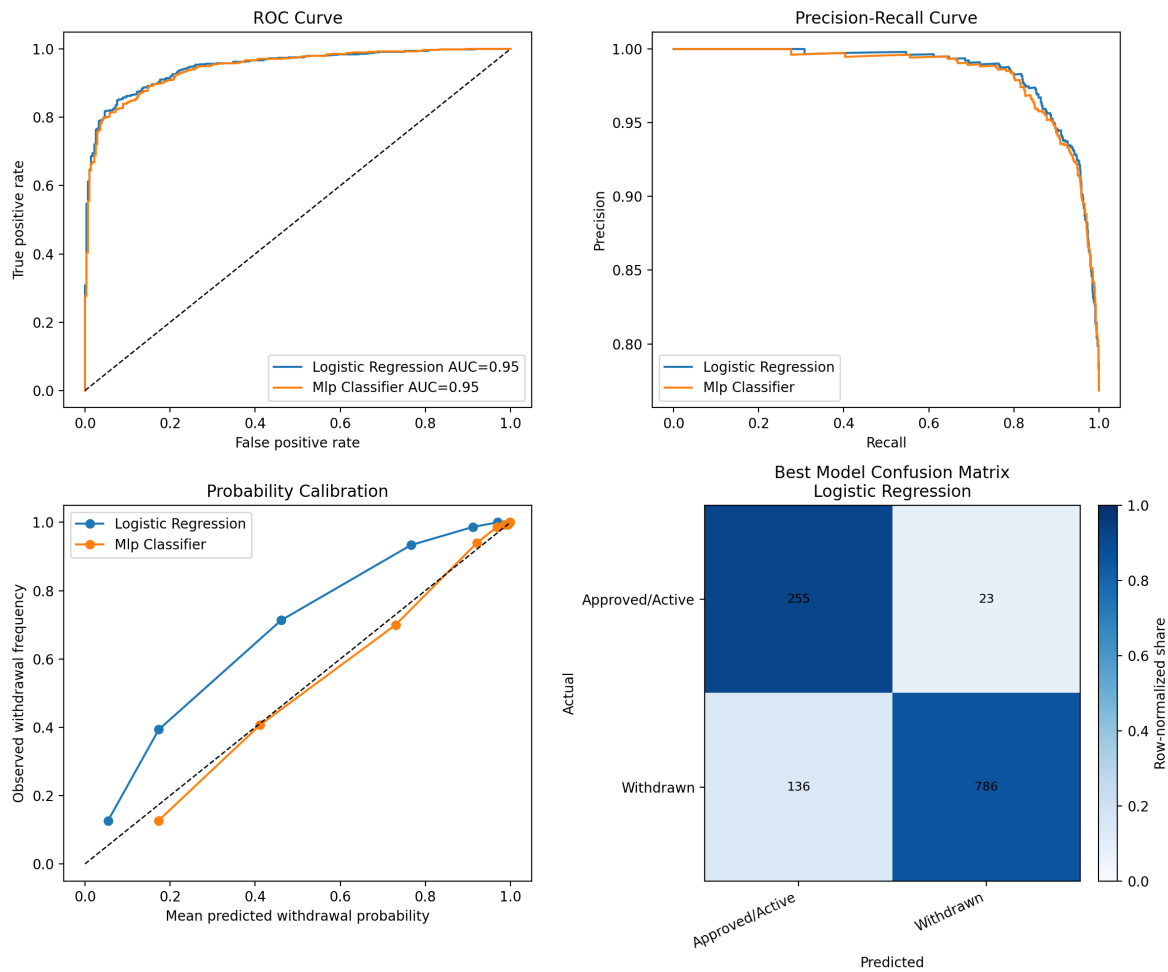


Figure 3: Withdrawal prediction diagnostics. Logistic regression and MLP models both separate withdrawn projects well.

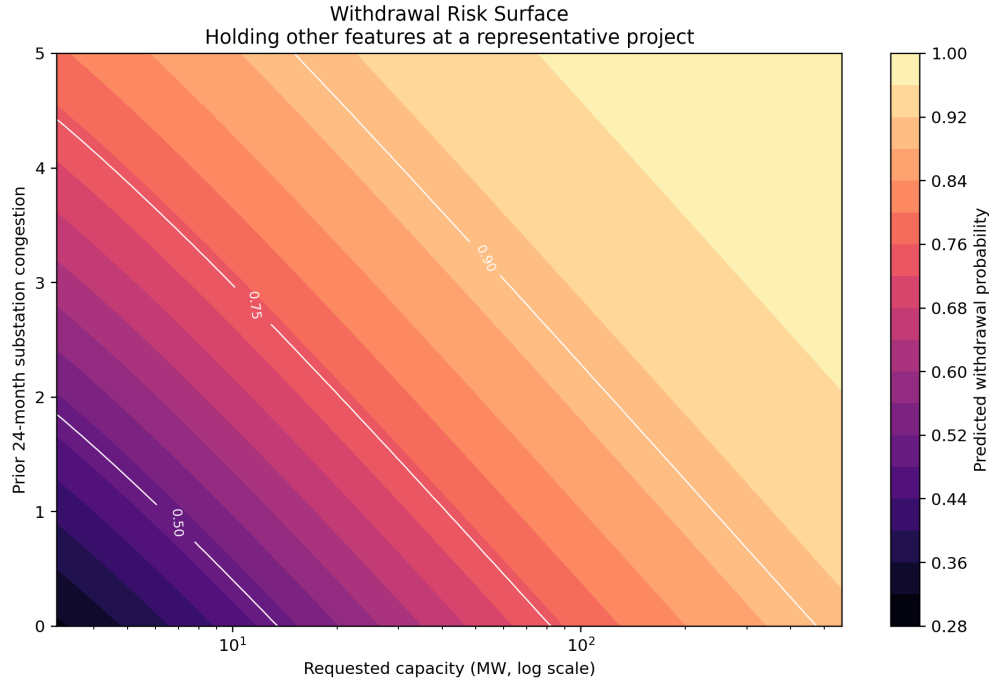


Figure 4: Predicted withdrawal risk surface as requested MW and prior 24-month substation congestion vary. Other features are held at representative values.

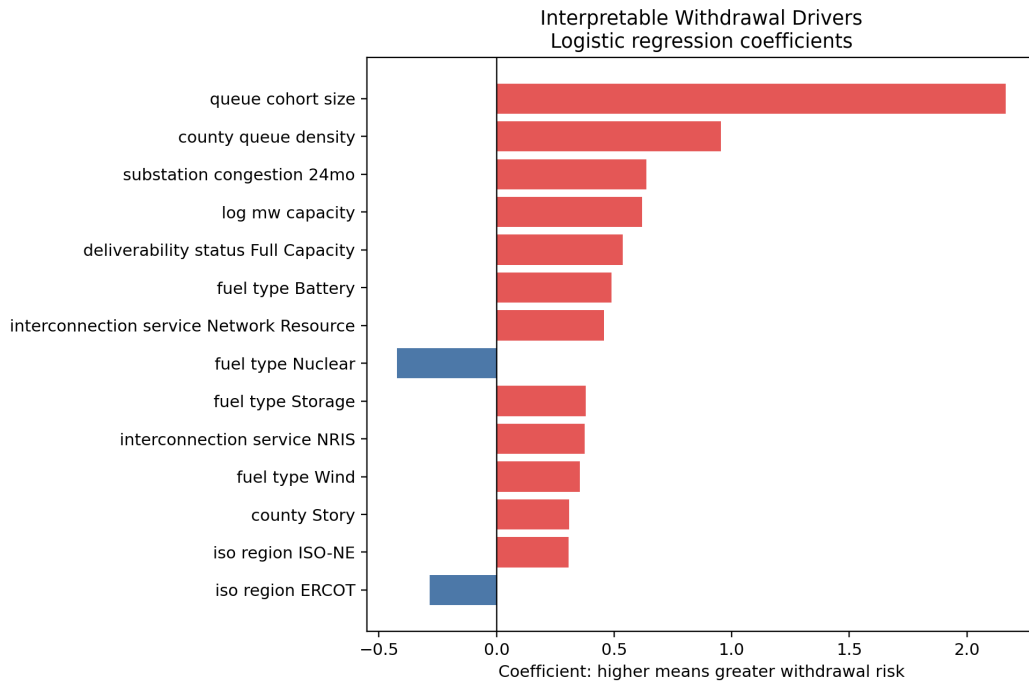


Figure 5: Largest logistic regression coefficients for withdrawal prediction. Positive coefficients increase predicted withdrawal risk; negative coefficients reduce it.

6.2 Study Timeline Prediction

Table 4: Study timeline prediction results.

Model	MAE (months)	R^2
Ridge regression	4.06	0.602
MLP regressor	4.13	0.590

Ridge regression slightly outperforms the MLP regressor, suggesting that much of the timeline signal in the reproducible dataset is captured by additive effects rather than requiring a more complex nonlinear model.

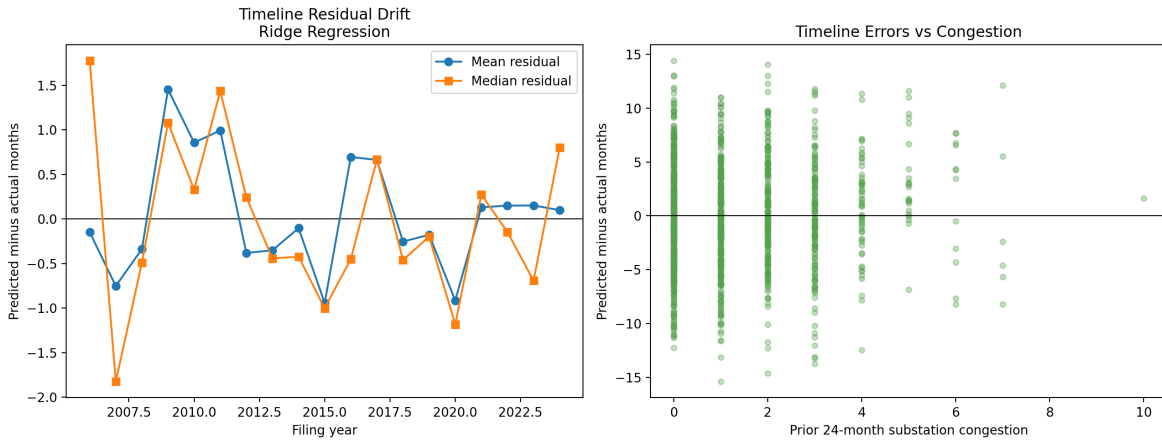


Figure 6: Timeline residual analysis. Residual drift by filing year and residuals against congestion reveal where the regression model over- or under-predicts study duration.

6.3 Cost-Bucket Classification and PINN Ablation

Table 5: Cost-bucket classification and PINN ablation results.

Model	Weighted F1	Accuracy	Implausibility rate
Unconstrained MLP	0.887	0.891	0.000
PINN monotonic MLP	0.879	0.886	0.000

The unconstrained model has slightly higher weighted F1 and accuracy in this run. The constrained model performs nearly as well while explicitly encoding the monotonic MW-cost prior. This is important to interpret carefully: the physics-informed term is not a guaranteed accuracy booster. Its value is to impose a domain-consistency prior, especially in sparse and noisy cost-label regions.

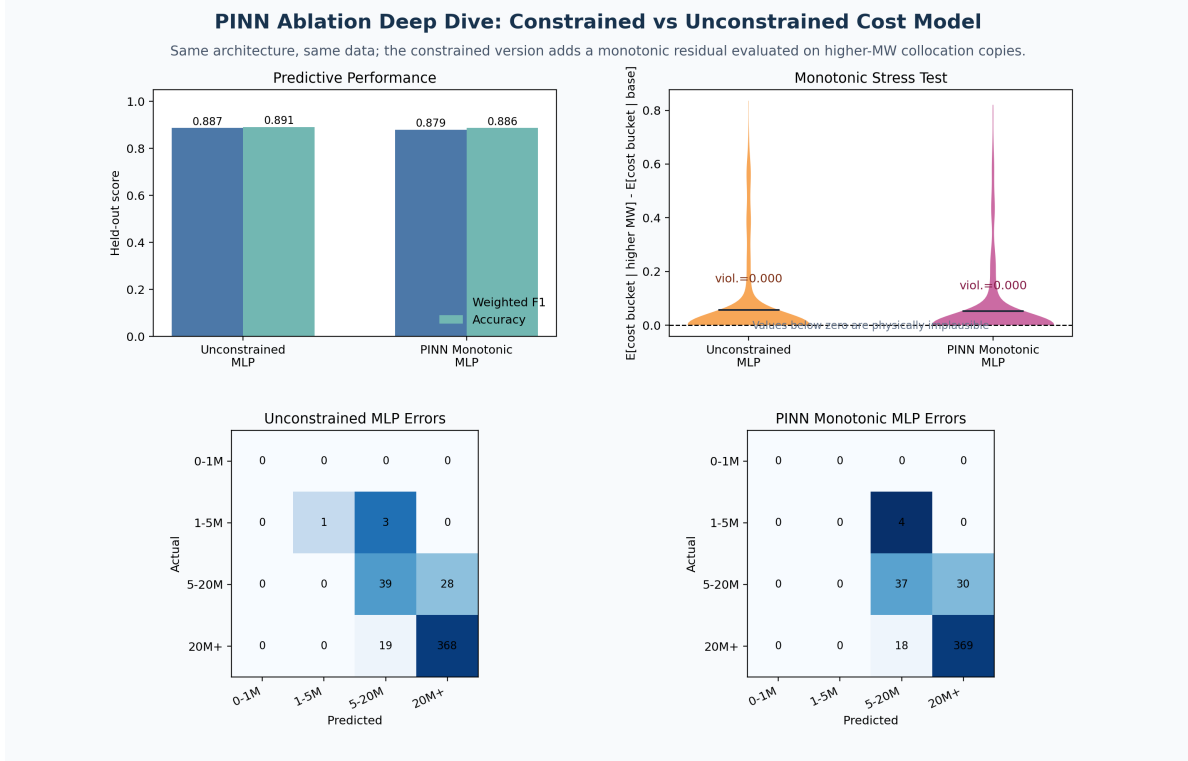


Figure 7: PINN ablation deep dive. The constrained model uses the same architecture as the unconstrained cost MLP but adds the monotonic residual from Equation 8.

6.4 Gaussian Process Cost Uncertainty

Table 6: Gaussian process cost uncertainty results.

Metric	Value
MAE on log cost	0.367
R^2 on log cost	0.782
90% prediction interval coverage	0.886
Mean interval width on log scale	1.516

The GPR model achieves R^2 of 0.782 on log cost and 88.6% empirical coverage for nominal 90% prediction intervals. The coverage is close to the target, indicating that the uncertainty estimates are reasonably calibrated for the reproducible dataset.

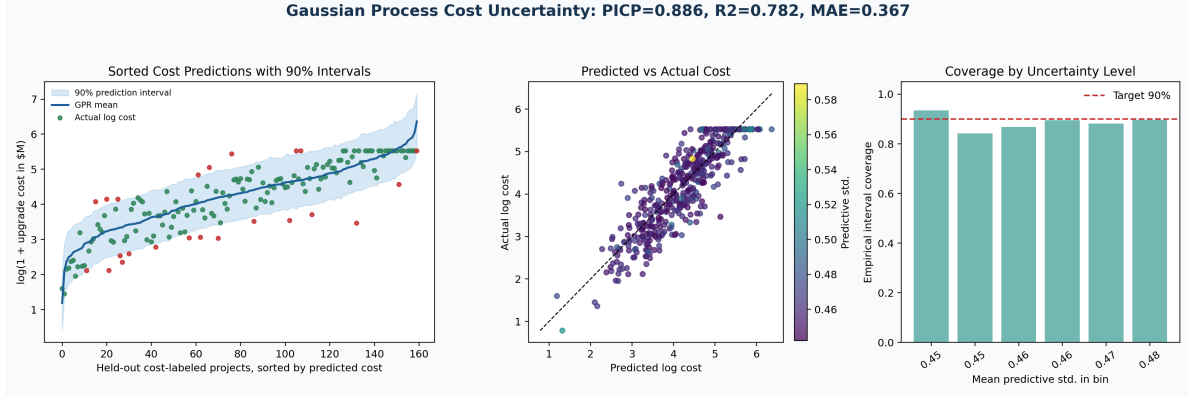


Figure 8: Gaussian process cost uncertainty diagnostics. The GPR model provides a posterior mean and 90% prediction interval for log upgrade cost.

7 Discussion

The first major insight is that queue records contain strong signal for withdrawal prediction. Both logistic regression and the MLP classifier achieved high AUC-ROC, and the MLP achieved high withdrawn-class F1. This suggests that project size, queue pressure, filing year, regional context, and local congestion can meaningfully distinguish projects that are likely to withdraw from those that are more likely to remain active or proceed.

The second insight is that interpretable baselines remain valuable. Ridge regression slightly outperformed the MLP regressor on timeline prediction, and logistic regression matched or exceeded the MLP on AUC for withdrawal prediction. These results do not weaken the project. Instead, they show that some queue outcomes are well explained by structured feature engineering and additive effects. In an engineering setting, interpretability is valuable because developers and regulators may prefer a transparent model over a more complex black box when performance is similar.

The third insight is that cost prediction is the most difficult and least complete task. Withdrawal and timeline labels are available for the full dataset, but network upgrade cost labels are available only for a subset. This changes the modeling problem. A cost model trained only on projects that reach later studies may not generalize perfectly to early withdrawals or projects with missing cost fields. GPR uncertainty and the PINN-style monotonic residual are therefore useful because they add caution and domain structure to the most label-constrained part of the project.

8 Limitations and Future Work

The largest limitation is data quality and completeness. Public queue exports differ across ISO/RTOs in field names, units, project identifiers, substation naming, queue status definitions, and update cadence. The pipeline normalizes these fields heuristically, but production-grade use would require much more careful data governance and validation against source documents.

The cost-label limitation is shown in Figure 9. Only 38.2% of records in the reproducible dataset have cost labels. More importantly, cost labels are not missing completely at random. Upgrade cost estimates are more likely to appear for projects that advanced far enough in the study process to receive facilities-study or network-upgrade estimates. This means the cost task is vulnerable to

selection bias.

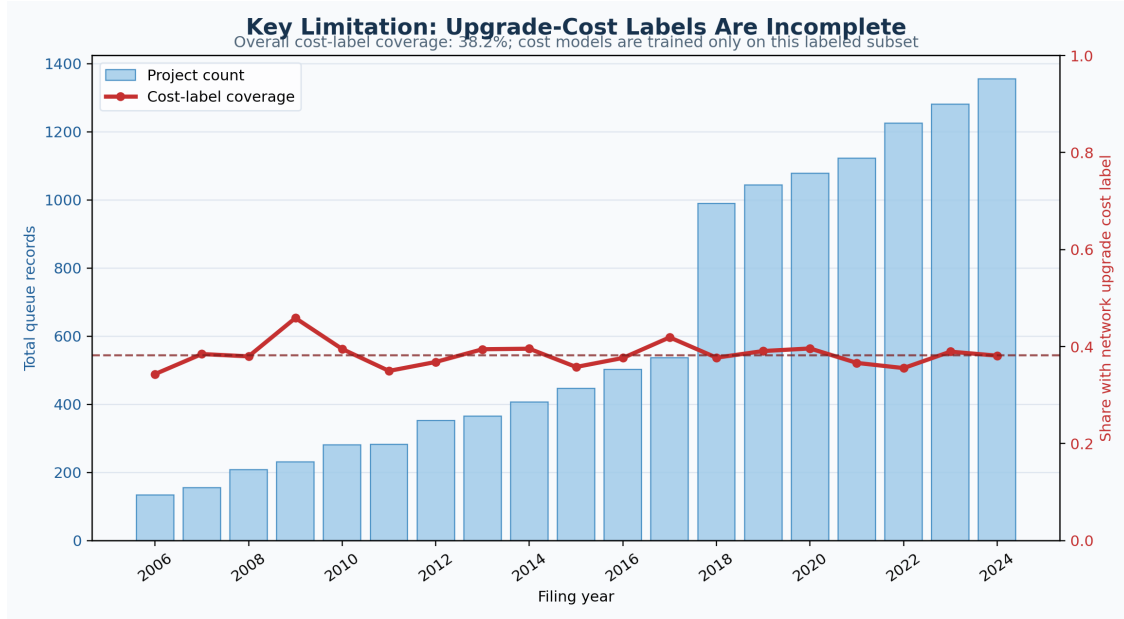


Figure 9: Cost-label limitation. Upgrade-cost labels are incomplete, so cost models are trained on a smaller and potentially biased subset of projects.

The second limitation is the simplified physics constraint. Actual network upgrade cost depends on deliverability requirements, thermal constraints, voltage constraints, short-circuit duty, affected-system studies, topology, and prior network upgrades. Requested MW is an important driver, but it is not sufficient to determine cost. The monotonic residual should therefore be interpreted as a useful regularizer, not a complete physical model.

The third limitation is Gaussian process scalability. Exact GPR training scales as $O(n^3)$ in the number of training examples. To keep the pipeline runnable, the GPR training subset is capped at 650 cost-labeled samples. A production implementation would need sparse Gaussian process approximations, inducing points, scalable kernels, or another calibrated uncertainty method.

Future work should replace the fallback/simplified data path with a fully cleaned and documented public-data integration across LBNL, MISO, PJM, CAISO, and other ISO/RTO exports. The model should also incorporate richer grid topology and study-context features: nearby transmission voltage, thermal headroom, line loading proxies, substation equipment age, prior network upgrades, deliverability study results, and affected-system flags. Cost-label modeling should explicitly handle missingness through semi-supervised learning, inverse-probability weighting, missingness models, or multi-task learning where withdrawal, timeline, and cost labels share a representation. Finally, the pipeline could be converted into an interactive developer-facing screening tool.

9 Conclusion

This project developed a complete machine-learning pipeline for early-stage interconnection queue intelligence. The system predicts withdrawal risk, study timeline, and network upgrade cost severity using project-level queue records and engineered grid-context features. Withdrawal prediction

was the strongest task, with AUC-ROC near 0.95 and withdrawn-class F1 above 0.90. Timeline prediction achieved approximately four months MAE. Cost-bucket prediction achieved weighted F1 near 0.89, and Gaussian process regression provided reasonably calibrated 90% intervals for log upgrade cost.

The project also demonstrated a principled adaptation of physics-informed neural networks. Rather than enforcing a PDE residual, the cost model adds a monotonic domain-residual term evaluated on higher-MW collocation copies. This residual encodes the grid-informed prior that increasing requested capacity should not reduce expected upgrade cost severity when context is fixed. The ablation shows that this constraint can be added while preserving similar predictive accuracy.

The main conclusion is that queue data can support useful early-stage risk screening, but the results should not be interpreted as replacements for formal interconnection studies. The model is best viewed as a decision-support layer: it can flag high-risk projects, identify uncertain cost estimates, and guide developers toward more informed siting decisions before committing substantial resources.

Statement of Individual Contributions

This project was completed independently by Eshana Jain. All components of the project, including project framing, literature review, data-source identification, feature engineering design, implementation of baseline models, implementation of neural-network models, implementation of the PINN-style monotonic residual, Gaussian process uncertainty modeling, evaluation, figure generation, presentation preparation, and final report writing, were the responsibility of the author.

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